

1. Administrative & Study Identification

Full Study Title: A Phase 1/2a, First-in-human, Randomized, Double-blinded, Placebo-controlled, Dose-finding Study in Healthy Volunteers and Participants with Sickle Cell Disease to Evaluate the Safety and Tolerability, Pharmacokinetics, Pharmacodynamics, pH and Food Effect, and Preliminary Efficacy of BMS-986470 **Short Title/Acronym:** A Study to Evaluate BMS-986470 in Healthy Volunteers and Participants With Sickle Cell Disease **Protocol Number:** CA230-1019 **NCT Number:** NCT06481306 **Posting ID:** 546554831253896981 **Sponsor Name:** Bristol Myers Squibb (Bristol-Myers Squibb Services Unlimited Company) **Investigational Product:** BMS-986470 (Oral HbF-activating cereblon E3 ligase modulator agent / dual molecular glue degrader) **Phase of Development:** Phase 1/Phase 2 (Phase 1/2a) **Trial Status:** Recruiting (Currently recruiting participants) **Disease Areas:** Anemia, Sickle Cell, Healthy Volunteers **Medical Monitor:** GSM-CT Representative (Bristol-Myers Squibb)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and tolerability, pharmacokinetics and pharmacodynamics, pH and food effect, and preliminary efficacy of BMS-986470 in healthy volunteers (Cohort A) and participants with sickle cell disease (Cohort B). **Secondary Objectives:** To determine the highest safe dose of BMS-986470 and evaluate if it successfully increases fetal hemoglobin (HbF) to improve SCD complications and reduce pain.

2.2 Background & Rationale Sickle cell disease (SCD) is a genetic disorder caused by a point mutation in the β -globin gene, producing sickle hemoglobin (HbS) that polymerizes under hypoxia, leading to rigid red blood cells and pain crises. The reactivation of fetal hemoglobin (HbF, $\alpha_2\gamma_2$) is a recognized disease modifier for SCD, where standard treatments are limited. BMS-986470 is an orally bioavailable agent that targets the transcriptional repressors of HbF (ZBTB7A and WIZ) for degradation, representing a novel mechanism of action to increase HbF levels and reduce disease complications.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blinded **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target **N**: 184 **Number of Sites**: 13 Locations (including sites in the US and UK) **Estimated Study Start**: 2024 **Estimated Primary Completion**: [TBD]

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. **Cohort A (Healthy Volunteers)**: Healthy adults with a body mass index (BMI) of 18.0 to 32.0 kg/m²; females must not be of childbearing potential (amenorrhoeic for ≥ 12 months without an alternative medical cause and with FSH ≥ 40 IU/L, or surgically sterile via hysterectomy, bilateral oophorectomy, or bilateral salpingectomy). No evidence of organ dysfunction or any clinically significant deviation from normal in physical examination, vital signs, ECG, or clinical laboratory assessments. 3. **Cohort B (SCD Participants)**: Documented diagnosis of SCD with genotype HbSS, HbS β 0-thalassemia, or HbS β +thalassemia; ≥ 4 vaso-occlusive crises (VOCs) within the previous 12 months or ≥ 2 VOCs within the last 6 months; Eastern Cooperative Oncology Group (ECOG) performance status of 0 or 1. 4. **Cohort B Laboratory Values**: Hemoglobin ≥ 5.5 and ≤ 12 g/dL (males) or ≥ 5.5 and ≤ 10.6 g/dL (females); Absolute neutrophil count $\geq 1500/\mu\text{L}$; Platelet count $\geq 100 \times 10^3/\mu\text{L}$; Absolute reticulocyte count $> 100 \times 10^3/\mu\text{L}$ or $> 50 \times 10^3/\mu\text{L}$ if taking hydroxyurea.

4.2 Exclusion Criteria 1. **Cohort A**: Any significant medical condition, or any major surgery or planned surgery (except GI surgery) within 12 weeks of the first study intervention administration. 2. **Cohort B (SCD Participants)**: > 6 severe VOCs (requiring ≥ 24 hours of hospital admission) within 12 months prior, or any VOC requiring ≥ 24 hours of hospital admission within 30 days prior to the first dose. 3. **Cohort B (SCD Participants)**: Any episode of acute chest syndrome within the 6 months prior to the first dose. 4. **Cohort B (SCD Participants)**: Creatinine clearance (CrCl) < 60 mL/min/1.72m² using the Chronic Kidney Disease Epidemiology (CKD-EPI) equation. 5. **Cohorts A and B**: Participant is receiving regularly scheduled RBC or platelet transfusions or has received an RBC transfusion within 28 days and a platelet transfusion within 14 days prior to starting treatment with BMS-986470. 6. **General**: Any condition or laboratory abnormality placing the participant at unacceptable risk. Prior bone marrow transplant (BMT).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Multiple dose-finding cohorts; randomized to BMS-986470 or BMS-986470 plus placebo. **Route of Administration**: Oral **Duration of Treatment**: Divided into screening, treatment, and follow-up stages.

5.2 Concomitant & Prohibited Therapy Permitted: Standard medications for the treatment of pain crises. Famotidine is included as a study intervention to test pH/food effects.
Prohibited: Prior bone marrow transplantation.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day -1	Informed Consent, Medical History, Genotype Confirmation, Baseline Labs
Baseline	Day 0	Randomization, First Dose, Safety Baseline Assessment
Interim	Treatment Stage	Safety Labs, PK/PD Blood Draws, HbF measurements, VOC Monitoring
End of Study	Follow-up Stage	Final Safety Follow-up, Efficacy Assessment (HbF%), Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Safety and tolerability evaluated through: 1. Number of participants with adverse events (AEs). 2. Number of participants with serious adverse events (SAEs). 3. Number of participants with AEs meeting protocol-defined Dose Limiting Toxicity (DLT) criteria. **Measurement Method:** Clinical assessment and standard AE reporting criteria.

7.2 Secondary & Exploratory Endpoints * **Pharmacokinetics (PK):** Maximum observed plasma concentration (Cmax), Area under the concentration-time curve (AUC), and Time of maximum observed plasma concentration (Tmax). * **Preliminary Efficacy:** Reactivation and increase in the percentage of fetal hemoglobin (HbF).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of AEs and dose-limiting toxicities (DLTs) to determine the maximum tolerated dose in the healthy volunteer cohort before escalating in the SCD cohort. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for any SAEs or hospitalizations. **Data Monitoring Committee (DMC):** Internal sponsor and independent monitoring to review safety signals during the dose-escalation phase.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants receiving at least one dose) and Efficacy Set (participants with baseline and post-baseline HbF measurements). **Sample Size Justification:** Target enrollment of 184 participants, based on

standard Phase 1/2a dose-escalation statistical models (e.g., 3+3 or similar adaptive design) to define safety profiles. **Handling of Missing Data:** Standard handling for early discontinuations due to dose-limiting toxicities.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring for source data verification. **Audit Trail:** eCRF locking and query resolution via standard clinical data management systems.

11. Study Identifiers & Governance

Primary ID: CA230-1019 **Registry Number:** NCT06481306 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** A Study to Evaluate BMS-986470 in Healthy Volunteers and Participants With Sickle Cell Disease **Official Regulatory Title:** A Phase 1/2a, First-in-human, Randomized, Double-blinded, Placebo-controlled, Dose-finding Study in Healthy Volunteers and Participants with Sickle Cell Disease to Evaluate the Safety and Tolerability, Pharmacokinetics, Pharmacodynamics, pH and Food Effect, and Preliminary Efficacy of BMS-986470

12. Clinical Framework (Sickle Cell Disease Specific)

Primary Condition: Sickle Cell Disease (SCD) **Diagnosis Threshold:** Documented HbSS, HbS β 0-thalassemia, or HbS β + -thalassemia genotypes **Medical Methodology:** Fetal Hemoglobin (HbF) Reactivation via CELMoD **Optimization Target:** Increase HbF levels and reduce the frequency of vaso-occlusive crises (VOCs)

13. Study Procedures & Methodology

Management Pathway: Dose-finding escalation (Healthy volunteers $\rightarrow$ SCD Participants) **Education Protocol (HCP):** Site initiation and dose-escalation toxicity management training **Education Protocol (Patient):** Informed consent detailing genetic therapy, oral dosing compliance, and VOC tracking **Quality Audit Mechanism:** Clinical site monitoring and independent safety review for DLTs

14. Observation & Follow-Up Schedule

Total Duration: Divided across screening, treatment, and extended follow-up stages. **Onsite Visit Cadence:** Frequent during the

